NARAYANA
IIT - JEE ACADEMY - INDIA

SEC:: SR. JEE MAINS APEX | GTM-24 (FN) | Date: 24.01.2025
Time: 3.00Hrs. | JEE MAINS -2025 | Max. Marks: 300 M

KEYSHEET

PHYSICS

1	2	3	4	5	6	7	8	9	10
D	B	D	D	C	D	B	B	A	B
11	12	13	14	15	16	17	18	19	20
A	D	B	A	A	C	A	D	D	B
21	22	23	24	25					
45	10	24	20	40					

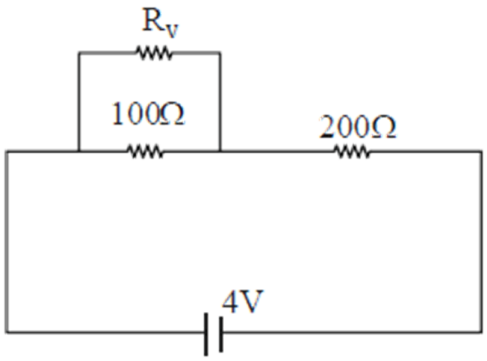
CHEMISTRY

26	27	28	29	30	31	32	33	34	35
D	D	D	B	B	B	C	C	C	A
36	37	38	39	40	41	42	43	44	45
B	B	C	A	B	C	A	A	C	B
46	47	48	49	50					
4	17	492	3	100					

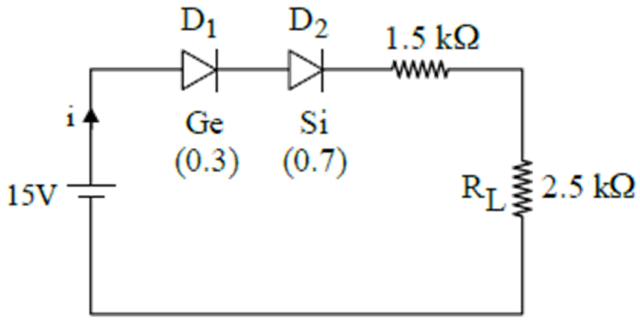
MATHEMATICS

51	52	53	54	55	56	57	58	59	60
D	B	D	A	C	C	A	B	B	C
61	62	63	64	65	66	67	68	69	70
C	D	A	D	A	B	A	B	D	B
71	72	73	74	75					
1575	10	9	960	5					

PHYSICS

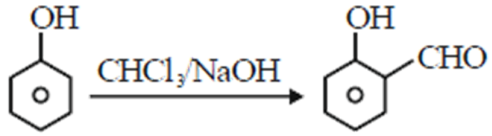
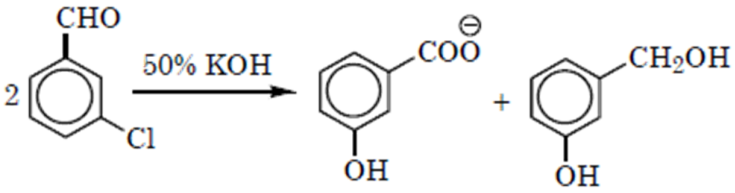
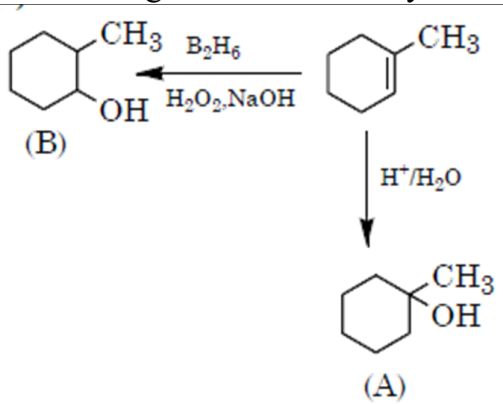
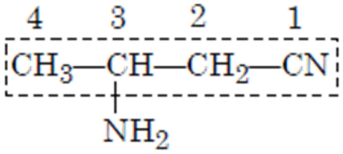
1.	$50\ V+S = 49S + S$ $S = 50\ (S - V)$ $.5 = 50\ (S - V)$ $S - V = \frac{0.5}{50} = \frac{1}{100} = 0.01\ mm$
2.	Work done again frictional force $= \mu N \times 10$ $= 0.1 \times 5 \times 10 = 5J$
3.	$K.E. = hf - \phi$ $\tan\theta = h$
4.	Conceptual
5.	$(X) \rightarrow (Y) + (Z) + (P)$ $M \quad M/3 \quad M/3 \quad M/3$ $\Delta Mc^2 = \frac{1}{2}\frac{M}{3}V^2 + \frac{1}{2}\frac{M}{3}V^2 + \frac{1}{2}\frac{M}{3}V^2$ $V = c\sqrt{\frac{2\Delta M}{M}}$
6.	 $\frac{R_v 100}{R_v + 100} = \frac{200}{3}$ $3R_v = 2R_v + 200$ $R_v = 200$
7.	Magnetic moment $= i\pi r^2$ $\mu = \frac{evr}{2}$ $\mu \propto \left(\frac{1}{n}\right) n^2$ $\mu \propto n$ $x = 1$

8.	Rising half to peak $t = T/6$ $t = \frac{2\pi}{6\omega} = \frac{\pi}{3\omega} = \frac{\pi}{300\pi} = \frac{1}{300} = 3.33\,ms$
9.	$\frac{dy}{dx} = \tan\theta = \frac{x}{2} = \mu = \frac{1}{2}$ $x = 1$, $y = 1/4$
10.	$p = \frac{E}{C} = \frac{6.48\times10^{5}}{3\times10^{8}} = 2.16\times10^{-3}$
11.	Intensity of emergent light $= \frac{I_0}{2}\cos^2 45° = \frac{I_0}{4}$
12.	$R_P = \frac{R_E}{3}$, $M_P = \frac{M_E}{6}$ $V_e = \sqrt{\frac{2GM_e}{R_e}}$ …(i) $V_P = \sqrt{\frac{2GM_P}{R_P}}$ …(ii) $\frac{V_e}{V_p} = \sqrt{2}$ $V_P = \frac{V_e}{\sqrt{2}} = \frac{11.2}{\sqrt{2}} = 7.9$ km/sec
13.	$\frac{2k\lambda q}{r} = m\omega^2 r$ $\omega^2 = \frac{2k\lambda q}{mr^2}$ $\left(\frac{2\pi}{T}\right)^2 = \frac{2k\lambda q}{mr^2}$ $T = 2\pi r\sqrt{\frac{m}{2k\lambda q}}$
14.	$m = k\,c^P\,G^{-1/2}\,h^{1/2}$ $M^1L^0T^0 = [LT^{-1}]^P[M^{-1}L^3T^{-2}]^{-1/2}[ML^2T^{-1}]^{1/2}$ By comparing $P = 1/2$

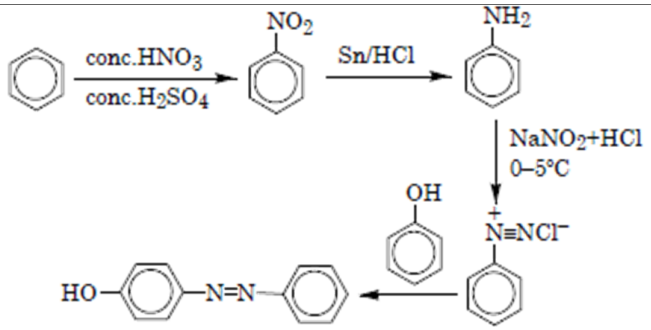
15.	 $i=\frac{14}{4}=3.5\text{mA}$ $V_L=iR_L=3.5\times2.5\text{ volt}$ $=8.75\text{volt}$
16.	$f_1=3,\quad f_2=5$ $n_1=3,\quad n_2=2$ $f_{mixture}=\frac{n_1f_1+n_2f_2}{n_1+n_2}=\frac{9+10}{f}=\frac{19}{5}$ $\gamma_{mixture}=1+\frac{2\times5}{19}=\frac{29}{19}=1.52$
17.	$a_A=a_B=a_C=\frac{F}{5+3+2}=\frac{80}{10}=8\text{m/s}^2$ A 5 kg $\rightarrow T_1$ $T_1=5\times8=40$ B $T_1\leftarrow$ 3 kg $\rightarrow T_2$ $T_2-T_1=3\times8\Rightarrow T_2=64$
18.	$\frac{v^2}{R}=W$ ….(i) $\frac{v^2}{\frac{1}{2}\left(\frac{R}{2}\right)}=W'$ ….(ii) From (i) & (ii), we get $W'=4W$
19.	Maxwell's equation

20.	Acceleration due to gravity $g' = \frac{g}{4}$ $T = 2\pi\sqrt{\frac{4\ell}{g}}$ $T = 2\pi\sqrt{\frac{4\times 4}{g}}$ $T = 2\pi\frac{4}{\pi} = 8s$
21.	$P_i = 2300 \times 5$ watt $P_0 = 2300 \times 5 \times 0.9 = 230 \times I_2$ $I_2 = 45A$
22.	$\rho\left(\frac{4}{3}\pi r^3\right)1000 = \frac{4}{3}\pi R^3\rho$ $R = 10r$ $E_1 = 1000 \times 4\pi r^2 \times S$ $E_2 = 4\pi\,(10r)^2\,S$ $\frac{E_1}{E_2} = \frac{10}{1},\ x = 10$
23.	$I_1\omega_1 + I_2\omega_2 = (I_1 + I_2)\omega_0$ (C.O.A.M.) gives $\omega_0 = 8$ rad/s $E_1 = \frac{1}{2}I_1\omega_1^2 + \frac{1}{2}I_2\omega_2^2 = 216J$ $E_2 = \frac{1}{2}(I_1 + I_2)\omega_0^2 = 192J$ $\therefore\ \Delta E = 24J$
24.	$\frac{1}{f+20} - \frac{1}{-(f+20)} = \frac{1}{f}$ $\frac{2}{f+20} = \frac{1}{f}\quad f = 20cm$ Or $x_1x_2 = f^2$ gives $f = 20$ cm
25.	$B = 4\times\frac{\mu_0 i}{4\pi\left(\frac{1}{2}\right)}\left(\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}}\right)$ $= 4\times10^{-7}\times5\times2\times\sqrt{2}$ $= 40\sqrt{2}\times10^{-7}$ T

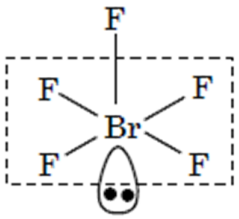
CHEMISTRY

26.	Different Extraction Different layers are formed which can be separated in funnel. (Theory based).
27.	
28.	**Statement – I:** Rate of $S_N2 \propto [R–X][Nu^-]$ S_N2 reaction is favoured by high concentration of nucleophile (Nu^-) & less crowding in the substrate molecule. **Statement – II:** Solvolysis follows S_N1 path. Both are correct Statements.
29.	**Sol.** Meta–chlorobenzaldehyde will undergo Cannizzaro reaction with 50% KOH to give m–chlorobenzoate ion and m–chlorobenzyl alcohol. 
30.	Due to lower Bond dissociation enthalpy of H_2Te it ionizes to give H^+ more easily than H_2S.
31.	
32.	 3–Aminobutanenitrile

33.



34. BrF_5



Square Pyramidal.

35. $K_2Cr_2O_7 \rightarrow Cr^{+6} \rightarrow$ No d – d transition
$KMnO_4 \rightarrow Mn^{7+} \rightarrow$ No d – d transition
} Charge transfer

36. Alkaline oxidative fusion of MnO_2:

$$2MnO_2 + 4OH^- + O_2 \rightarrow 2MnO_4^{2-} + 2H_2O$$

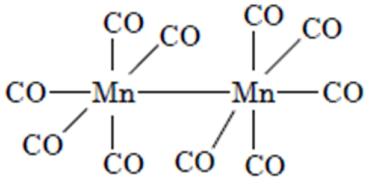
Electrolytic oxidation of MnO_4^{2-} in alkaline medium.

$$MnO_4^{2-} \rightarrow MnO_4^- + e^-$$

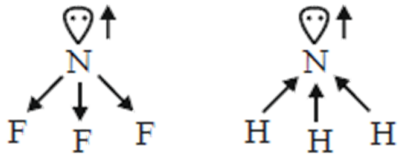
37. Mole fraction of $C = \dfrac{n_C}{n_A + n_B + n_C}$

38. Chemical reactivity of elements decreases along the period therefore statement – I is false.
Group – 1 elements from basic nature oxides while group – 17 elements form acidic oxides therefore statement – II is true.

39. $Mn_2(CO)_{10}$



Octahedral around Mn

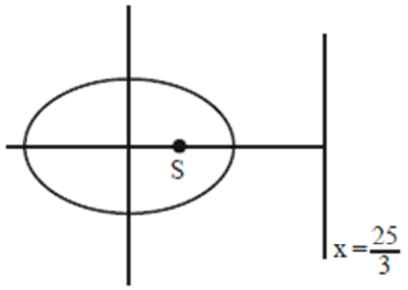
40.	
41.	More no. of hyperconjugable Hydrogens, more stable is the carbocations.
42.	ΔT_f is maximum when i × m is maximum. 1) $m_1 = \frac{180}{60} = 3$, $i = 1 + \alpha$ Hence $\Delta T_f = (1+\alpha)\cdot k_f = 3 \times 1.86 = 5.58\,^{\circ}C\ (\alpha << 1)$ 2) $m_2 = \frac{180}{60} = 3, i = 0.5, \Delta T_f = \frac{3}{2} \times k_f' = 7.68^{\circ}C$ 3) $m_3 = \frac{180}{122} = 1.48, i = 0.5, \Delta T_f = \frac{1.48}{2} \times k_f' = 3.8^{\circ}C$ 4) $m_4 = \frac{180}{180} = 1, i = 1, \Delta T_f = 1 \cdot k_f' = 1.86^{\circ}C$ As per NCERT, $k_f'(H_2O) = 1.86\ k \cdot kg\ mol^{-1}$ $k_f'(\text{Benzene}) = 5.12\ k \cdot kg\ mol^{-1}$
43.	**Sol.** $CrO_2Cl_2 + 4NaOH \rightarrow \underset{(A)}{Na_2CrO_4} + 2NaCl + 2H_2O$ $Na_2CrO_4 + 2H_2O_2 + 2HCl \rightarrow \underset{(B)}{CrO_5} + \underset{\text{Missing from balanced eqaution}}{2NaCl} + 3H_2O$
44.	On moving down the group, bond strength of M–H bond decreases, which reduces the thermal stability but increases reducing nature of hydrides, hence A and B are correct statements.
45.	Higher the value of ⊕ve SRP (Std. reduction potential) more is tendency to undergo reduction, so better is oxidising power of reactant. Hence, ox. Power:- $BrO_4^- > IO_4^- > ClO_4^-$

46.	$cis-[Cr(ox)_2Cl_2]^{3-} \rightarrow$ can show optical isomerism (no POS & COS) $[Co(en)_3]^{3+} \rightarrow$ can show (no POS & COS) $cis-[Pt(en)_2Cl_2]^{2+} \rightarrow$ can show (no POS & COS) $cis-[Co(en)_2Cl_2]^{+} \rightarrow$ can show (no POS & COS) $trans-[Pt(en)_2Cl_2]^{2+} \rightarrow$ can't show (contains POS & COS) $trans-[Cr(ox)_2Cl_2]^{3-} \rightarrow$ can't show (contains POS & COS)
47.	Rate of reaction (ROR) $= -\frac{1}{2}\frac{\Delta[N_2O_5]}{\Delta t} = \frac{1}{4}\frac{[NO_2]}{\Delta t} = \frac{\Delta[O_2]}{\Delta t}$ $ROR = -\frac{1}{2}\frac{\Delta[N_2O_5]}{\Delta t} = -\frac{1}{2}\frac{(2.75-3)}{30}\,mol\,L^{-1}\,min^{-1}$ $ROR = -\frac{1}{2}\frac{(-0.25)}{30}\,mol\,L^{-1}\,min^{-1}$ $ROR = \frac{1}{240}\,mol\,L^{-1}\,min^{-1}$ Rate of formation of $NO_2 = \frac{\Delta[NO_2]}{\Delta t} = 4 \times ROR$ $= \frac{4}{240} = 16.66\times10^{-3}\,molL^{-1}\,min^{-1} \simeq 17\times10^{-3}$.
48.	$2Fe_{(s)} + \frac{3}{2}O_{2(g)} \rightarrow Fe_2O_{3(s)}, \Delta H° = -822\,kJ/mol$ ……(1) $C_{(s)} + \frac{1}{2}O_{2(g)} \rightarrow CO_{(g)}, \Delta H° = -110\,kJ/mol$ ……(2) $3C_{(s)} + Fe_2O_{3(s)} \rightarrow 2Fe_{(s)} + 3CO_{(g)},\ \Delta H_3 = ?$ $(3) = 3 \times (2) - (1)$ $\Delta H_3 = 3 \times \Delta H_2 - \Delta H_1$ $= 3(-110) + 822$ $= 492$ kJ/mole
49.	a. RNA is regarded as the reserve of genetic information. (False) b. DNA molecule self-duplicates during cell division. (True)

	c. DNA synthesizes proteins in the cell. (False) d. The message for the synthesis of particular proteins is present in DNA. (True) d. Identical DNA strands are transferred to daughter cells. (True)
50.	1M Benzoic acid + 1M Sodium Benzoate (V_a ml) (V_s ml) Millimole $V_a \times 1$ $V_s \times 1$ pH = 4.5 $pH = pka + \log\frac{[salt]}{[acid]}$ $4.5 = 4.2 + \log\left(\frac{V_s}{V_a}\right)$ $\frac{V_s}{V_a} = 2$ .....(1) $V_s + V_a = 300$ ..... (2) $V_a = 100$ ml

MATHS

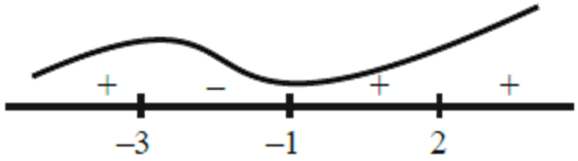
51.	$\begin{vmatrix}1 & 1 & 1\\ 1 & 2 & \lambda^2\\ 1 & 3 & \lambda\end{vmatrix}=0$ $\Rightarrow 2\lambda^2-\lambda-1=0$ $\lambda=1,\ -\frac{1}{2}$ $\begin{vmatrix}1 & 1 & 5\\ 2 & \lambda^2 & 9\\ 3 & \lambda & \mu\end{vmatrix}=0\Rightarrow \mu=13$ Infinite solution $\lambda=1$ & $\mu=13$ For unique sol^n $\lambda\neq 1$ For no sol^n $\lambda=1$ & $\mu\neq 13$ If $\lambda\neq 1$ and $\mu\neq 13$ Considering the case when $\lambda=-\frac{1}{2}$ and $\mu\neq 13$ this will generate no solution case
52.	$3\sin\alpha\cos\beta+3\sin\beta\cos\alpha$ $=2\sin\alpha\cos\beta-2\sin\beta\cos\alpha$ $5\sin\beta\cos\alpha=-\sin\alpha\cos\beta$ $\tan\beta=-\frac{1}{5}\tan\alpha$ $\tan\alpha=-5\tan\beta$

53.	$A = (10, 0)$ $B = \left(0, \frac{50}{7}\right)$ $\Big\}$ $P = (3, 5)$  $ae = 3$ $\frac{a}{e} = \frac{25}{3}$ $a = 5$ $b = 4$ Length of $LR = \frac{2b^2}{a} = \frac{32}{5}$
54.	$\lvert\vec{b}\rvert^2 = 6$; $\lvert\vec{a}\rvert\lvert\vec{b}\rvert\cos\theta = 3\sqrt{2}$ $\lvert\vec{a}\rvert^2 \lvert\vec{b}\rvert^2 \cos^2\theta = 18$ $\lvert\vec{a}\rvert^2 = 6$ Also $1 + \alpha^2 + \beta^2 = 6$ $\alpha^2 + \beta^2 = 5$ to find $(\alpha^2 + \beta^2)\lvert\vec{a}\rvert^2\lvert\vec{b}\rvert^2 \sin^2\theta$ $= (5)(6)(6)\left(\frac{1}{2}\right)$ $= 90$

55. **Sol.** $f'(x) = (x+3)^2 \cdot 3(x-2)^2 + (x-2)^3 2(x+3)$

$= 5(x+3)(x-2)^2(x+1)$

$f'(x) = 0, x = -3, -1, 2$



$f(-4) = -216$

$f(-3) = 0, f(4) = 49 \times 8 = 392$

$M = 392, m = -216$

$M - m = 392 + 216 = 608$

Ans = '3'

56. 1st GP $\Rightarrow t_1 = a, t_3 = b = ar^2 \Rightarrow r^2 = \dfrac{b}{a}$

$$t_{11} = ar^{10} = a(r^2)^5 = a \cdot \left(\frac{b}{a}\right)^5$$

2nd G.P. $\Rightarrow T_1 = a, T_5 = ar^4 = b$

$$\Rightarrow r^4 = \left(\frac{b}{a}\right) \Rightarrow r = \left(\frac{b}{a}\right)^{1/4}$$

$$T_p = ar^{p-1} = a\left(\frac{b}{a}\right)^{\frac{p-1}{4}}$$

$$t_{11} = T_p \Rightarrow a\left(\frac{b}{a}\right)^5 = a\left(\frac{b}{a}\right)^{\frac{p-1}{4}}$$

$$\Rightarrow 5 = \frac{p-1}{4} \Rightarrow p = 21$$

57. Cocus of point P(x, y) whose distance from

Gives

X + 2y + 7 = 0 & 2x – y + 8 = 0 are equal is

$$\frac{x+2y+7}{\sqrt{5}} = \pm\frac{2x-y+8}{\sqrt{5}}$$

$(x+2y+7)^2 - (2x-y+8)^2 = 0$

	Combined equation of lines $(x-3y+1)(3x+y+15)=0$ $3x^2-3y^2-8xy+18x-44y+15=0$ $x^2-y^2-\frac{8}{3}xy+6x-\frac{44}{3}y+5=0$ $x^2-y^2+2h\,xy+2gx\,2+2fy+c=0$ $h=\frac{4}{3},\ g=3,\ f=-\frac{22}{3},\ c=5$ $g+c+h-f=3+5-\frac{4}{3}+\frac{22}{3}=8+6=14$
58.	$\lvert\vec{b}\rvert=1$ & $\lvert\vec{b}\times\vec{a}\rvert=2$ $\left(\vec{b}\times\vec{a}\right)\cdot\vec{b}=\vec{b}\cdot\left(\vec{b}\times\vec{a}\right)=0$ $\left\lvert(\vec{b}\times\vec{a})-\vec{b}\right\rvert^2=\left\lvert\vec{b}\times\vec{a}\right\rvert^2+\left\lvert\vec{b}\right\rvert^2$ $=4+1=5$
59.	$y=f(x)\Rightarrow\frac{dy}{dx}=f'(x)$ $\left.\frac{dy}{dx}\right)_{(1,f(1))}=f'(1)=\tan\frac{\pi}{6}=\frac{1}{\sqrt{3}}\Rightarrow f'(1)=\frac{1}{\sqrt{3}}$ $\left.\frac{dy}{dx}\right)_{(3,f(3))}=f'(3)=\tan\frac{\pi}{4}=1\Rightarrow f'(3)=1$ $27\int_1^3\left((f'(t))^2+1\right)f''(t)dt=\alpha+\beta\sqrt{3}$ $I=\int_1^3\left((f'(t))^2+1\right)f''(t)dt$ $f'(t)=z\Rightarrow f''(t)\,dt=dz$ $z=f'(3)=1$ $z=f'(1)=\frac{1}{\sqrt{3}}$

$$I=\int_{1/\sqrt{3}}^{1}(z^2+1)dz=\left(\frac{z^3}{3}+z\right)_{1/\sqrt{3}}^{1}$$

$$=\left(\frac{1}{3}+1\right)-\left(\frac{1}{3}\cdot\frac{1}{3\sqrt{3}}+\frac{1}{\sqrt{3}}\right)$$

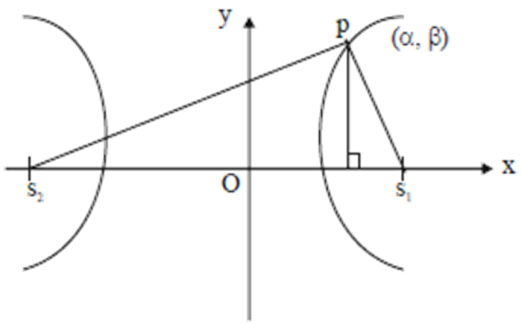
$$=\frac{4}{3}-\frac{10}{9\sqrt{3}}=\frac{4}{3}-\frac{10}{27}\sqrt{3}$$

$$\alpha+\beta\sqrt{3}=27\left(\frac{4}{3}-\frac{10}{27}\sqrt{3}\right)=36-10\sqrt{3}$$

$\alpha=36,\ \beta=-10$

$\alpha+\beta=36-10=26$

60.



$$\frac{x^2}{9}-\frac{y^2}{4}=1$$

$a^2=9,\ b^2=4$

$$b^2=a^2(e^2-1)\Rightarrow e^2=1+\frac{b^2}{a^2}$$

$$e^2=1+\frac{4}{9}=\frac{13}{9}$$

$$e=\frac{\sqrt{13}}{3}\Rightarrow s_1s_2=2ae=2\times3\times\sqrt{\frac{13}{3}}=2\sqrt{13}$$

$$\text{Area of } \Delta PS_1S_2=\frac{1}{2}\times\beta\times s_1s_2=2\sqrt{13}$$

$$\Rightarrow\frac{1}{2}\times\beta\times(2\sqrt{13})=2\sqrt{13}\Rightarrow\beta=2$$

$$\frac{\alpha^2}{9}-\frac{\beta^2}{4}=1\Rightarrow\frac{\alpha^2}{9}-1=1\Rightarrow\alpha^2=18\Rightarrow\alpha=3\sqrt{2}$$

Distance of P from origin $=\sqrt{\alpha^2+\beta^2}$

$$=\sqrt{18+4}=\sqrt{22}$$

61.

E_1 : A is selected

A	B
3W	3W
7R	2R

E_2 : B is selected

E : white ball is drawn

$P(E_1/E) =$

$$\frac{P(E).P(E/E_1)}{P(E_1).P(E/E_1)+P(E_2).P(E/E_2)}=\frac{\frac{1}{2}\times\frac{3}{10}}{\frac{1}{2}\times\frac{3}{10}+\frac{1}{2}\times\frac{3}{5}}$$

$$=\frac{3}{3+6}=\frac{1}{3}$$

62.

$f(x) = ae^{2x} + be^x + cx$ $\quad f(0) = -1$

$a + b = -1$

$f'(x) = 2ae^{2x} + be^x + c$ $\quad f'(\ln 2) = 21$

$8a + 2b + c = 21$

$$\int_0^{\ln 4}(ae^{2x}+be^x)dx=\frac{39}{2}$$

$$\left[\frac{ae^{2x}}{2}+be^x\right]_0^{\ln 4}=\frac{39}{2} \Rightarrow 8a+4b-\frac{a}{2}-b=\frac{39}{2}$$

$15a + 6b = 39$

$15a - 6a - 6 = 39$

$9a = 45 \Rightarrow a = 5$

$b = -6$

$c = 21 - 40 + 12 = -7$

$a + b + c - 8$

$|a + b + c| = 8$

63.

$L_1 \perp L_2$ $\quad L_3 \perp L_1,\ L_2$

$3 - 1 + 2P = 0$

$P = -1$

$$\begin{vmatrix}\hat{i} & \hat{j} & \hat{k}\\ 1 & -1 & 2\\ 3 & 1 & -1\end{vmatrix}=-\hat{i}+7\hat{j}+4\hat{k}$$

$\therefore$ $(-\delta, 7\delta, 4\delta)$ will lie on L_3

For $\delta = 1$ the point will be (-1, 7, 4)

64.

f is continuous $\quad f'(x) = 2x + 3 \ , \ k < 1$

$\therefore \quad 4 + a = b + 2 \qquad\qquad b \quad , \ x > 1$

$a = b - 2 \qquad$ f is differentiable

$\therefore \quad b = 5$

$\therefore \quad a = 3$

$$\int_{-2}^{1}(x^2 + 3x + 3)dx + \int_{1}^{2}(5x + 2)dx$$

$$= \left[\frac{x^3}{3} + \frac{3x^2}{2} + 3x\right]_{-2}^{1} + \left[\frac{5x^2}{2} + 2x\right]_{1}^{2}$$

$$= \left(\frac{1}{3} + \frac{3}{2} + 3\right) - \left(\frac{-8}{3} + 6 - 6\right) + \left(10 + 4 - \frac{5}{2} - 2\right)$$

$$= \quad 6 + \frac{3}{2} + 12 - \frac{5}{2} = 17$$

65.

$$f\left(\frac{x}{y}\right) = \frac{f(x)}{f(y)} \qquad f'(1) = 2024 \qquad f(1) = 1$$

Partially differentiating w. r. t. x

$$f'\left(\frac{x}{y}\right).\frac{1}{y} = \frac{1}{f(y)}f'(x)$$

$y \to x$

$$f'(1).\frac{1}{x} = \frac{f'(x)}{f(x)}$$

$2024f(x) = xf'(x) \ \Rightarrow \ xf'(x) - 2024\, f(x) = 0$

66.

Sol. $z^{1985} + z^{100} + 1 = 0 \quad \& \quad z^3 + 2z^2 + 2z + 1 = 0$

$(z + 1)\,(z^2 - z + 1) + 2z(z + 1) = 0$

$(z + 1)\ (z^2 + z + 1) = 0$

$\Rightarrow \quad z = -1 \ , \quad z = w, w^2$

Now putting $z = -1$ not satisfy

Now put $z = w$

$\Rightarrow \quad w^{1985} + w^{100} + 1$

$\Rightarrow \quad w^2 + w + 1 = 0$

Also, $z = w^2$

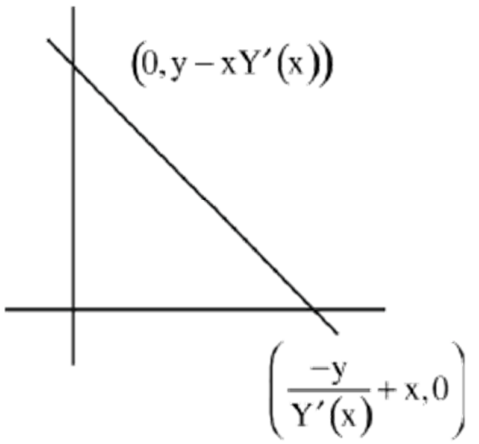
$\Rightarrow \quad w^{3970} + w^{200} + 1$

$\Rightarrow \quad w + w^2 + 1 = 0$

Two common root

67.

$$A=\frac{1}{2}\left(\frac{-y}{Y'(x)}+x\right)(y-xY/x)\ =\frac{-y^2}{2Y'(x)}+1$$

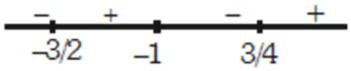


$$\Rightarrow\ (-y+xY'(x))(y-xY'(x))=-y^2+2Y'(x)$$

68.

$$\frac{2x+3}{4x^2+x-3}>0 \text{ and } -1\le\frac{2x-1}{x+2}\le 1$$

$$\frac{2\times+3}{(4x-3)(x+1)}>0 \quad \frac{3x+1}{x+2}\ge 0 \ \&\ \frac{x-3}{x+2}\le 0$$



$$(-\infty,-2)\cup\left[\frac{-1}{3},\infty\right) \quad .....(1)$$

$$(-2,3] \quad .....(2)$$

$$\left[\frac{-1}{3},3\right] \quad .....(3) \quad (1)\cap(2)\cap(3)$$

$$\left(\frac{3}{4},3\right]$$

$$\alpha=\frac{3}{4}\quad \beta=3$$

$$5\beta-4\alpha\ =\ 15-3=12$$

69.

$$f(x)=\frac{x}{(1+x^4)^{1/4}}$$

$$fof(x)=\frac{f(x)}{(1+f(x)^4)^{1/4}}=\frac{\frac{x}{(1+x^4)^{1/4}}}{\left(1+\frac{x^4}{1+x^4}\right)^{1/4}}=\frac{x}{(1+2x^4)^{1/4}}$$

$$f(f(f(f(x))))=\frac{x}{(1+4x^4)^{1/4}}$$

$$18\int_0^{\sqrt{2\sqrt{5}}}\frac{x^3}{(1+4x^4)^{1/4}}dx$$

Let $1+4x^4=t^4$

$16x^3\,dx=4t^3\,dt$

$$\frac{18}{4}\int_1^3\frac{t^3dt}{t}$$

$$=\frac{9}{2}\left(\frac{t^3}{3}\right)_1^3$$

$$=\frac{3}{2}[26]=39$$

70.

$$x\sin\theta=y\sin\left(\theta+\frac{2\pi}{3}\right)=z\sin\left(\theta+\frac{4\pi}{3}\right)\neq 0$$

$\Rightarrow x,y,z\neq 0$

Also,

$$\sin\theta+\sin\left(\theta+\frac{2\pi}{3}\right)+\sin\left(\theta+\frac{4\pi}{3}\right)=0\ \forall\ \theta\in R$$

$$\Rightarrow\frac{1}{x}+\frac{1}{y}+\frac{1}{z}=0$$

$\Rightarrow xy+yz+zx=0$

(i) Trace (R) = x + y + z

If x + y + z = 0 and x y + yz + zx = 0

$\Rightarrow x=y=z=0$

Statement (i) is False

(ii) Adj(Adj(R)) = |R| R

Trace (Adj(Adj(R)))

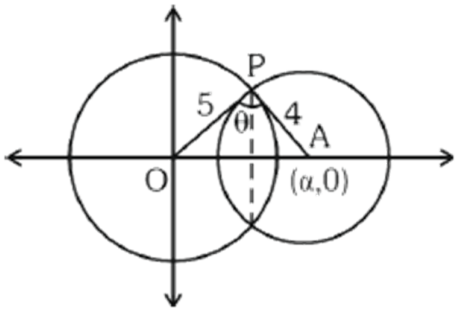
= xyz (x + y + z) ≠ 0

Statement (ii) is also False

71.

$C_1 : x^2 + y^2 = 25$, $C_2 : (x-\alpha)^2 + y^2 = 16$

$5 < \alpha < 9$



$\theta = \sin^{-1}\left(\frac{\sqrt{63}}{8}\right)$

$\sin\theta = \frac{\sqrt{63}}{8}$

Area of $\Delta OAP = \frac{1}{2} \times \alpha\left(\frac{\beta}{2}\right) = \frac{1}{2} \times 5 \times 4\sin\theta$

$\alpha\beta = 40 \times \frac{\sqrt{63}}{8}$

$\alpha\beta = 5 \times \sqrt{63}$

$(\alpha\beta)^2 = 25 \times 63 = 1575$

72.

$$\alpha = \sum_{k=0}^{n} \frac{{}^{n}C_k \cdot {}^{n}C_k}{k+1} \cdot \frac{n+1}{n+1}$$

$$= \frac{1}{n+1} \sum_{k=0}^{n} {}^{n+1}C_{k+1} \cdot {}^{n}C_{n-k}$$

$$\alpha = \frac{1}{n+1} \cdot {}^{2n+1}C_{n+1}$$

$$\beta = \sum_{k=0}^{n-1} {}^{n}C_k \cdot \frac{{}^{n}C_{k+1}}{k+2} \frac{n+1}{n+1}$$

$$\frac{1}{n+1} \sum_{k=0}^{n-1} {}^{n}C_{n-k} \cdot {}^{n+1}C_{k+2}$$

$$= \frac{1}{n+1} \cdot {}^{2n+1}C_{n+2}$$

$$\frac{\beta}{\alpha} = \frac{{}^{2n+1}C_{n+2}}{{}^{2n+1}C_{n+1}} = \frac{2n+1-(n+2)+1}{n+2}$$

$$\frac{\beta}{\alpha} = \frac{n}{n+2} = \frac{5}{6}$$

$$n = 10$$

73.

$S_n = 3 + 7 + 11 + \ldots\ldots$ n terms

$$= \frac{n}{2}\left(6 + (n-1)4\right) = 3n + 2n^2 - 2n$$

$$= 2n^2 + n$$

$$\sum_{k=1}^{n} S_k = 2\sum_{k=1}^{n} K^2 + \sum_{k=1}^{n} K$$

$$= 2 \cdot \frac{n(n+1)(2n+1)}{6} + \frac{n(n+1)}{2}$$

$$= n(n+1)\left[\frac{2n+1}{3} + \frac{1}{2}\right]$$

$$= \frac{n(n+1)(4n+5)}{6}$$

$$\Rightarrow \quad 40 < \frac{6}{n(n+1)} \sum_{k=1}^{n} S_k < 42$$

$$40 < 4n + 5 < 42$$

$$35 < 4n < 37$$

$$n = 9$$

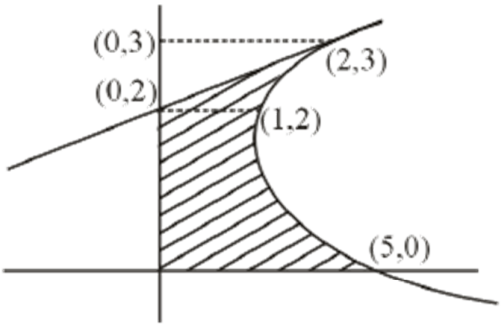
74.	Total number of relation both symmetric and reflexive = $2^{\frac{n^2-n}{2}}$ Total number of symmetric relation = $2^{\left(\frac{n^2+n}{2}\right)}$ $\Rightarrow$ Then number of symmetric relation which are not reflexive $\Rightarrow 2^{\frac{n(n+1)}{2}} - 2^{\frac{n(n-1)}{2}}$ $\Rightarrow 2^{10} - 2^6$ $\Rightarrow 1024 - 64$ $= 960$

75.

Solving the equations

$(y-2)^2 = x-1$ and $x-2y+4=0$

$X = 2(y-2)$



$\frac{x^2}{4} = x-1$

$x^2 - 4x + 4 = 0$

$(x-2)^2 = 0$

$x = 2$

Exclose area (w.r.t. y-axis) $= \int_0^3 x\,dy$ – Area of Δ

$= \int_0^3 \left((y-2)^2+1\right)dy - \frac{1}{2}\times 1\times 2$

$= \int_0^3 \left(y^2-4y+5\right)dy - 1$

$= \left[\frac{y^3}{3} - 2y^2 + 5y\right]_0^3 - 1$

$= 9 - 18 + 15 - 1 = 5$